Unified Business Management Platform: Problem Statement

1. Background and Problem

Small and growing service businesses lose revenue because inbound leads are answered late or never. A lead arrives by web form, chat or phone; the employee it was meant for is absent, busy or missed the notification; nobody notices; the customer goes to a competitor. The company cannot even see how often this happens, because customer data, conversations, payments, appointments and employee activity are scattered across separate tools that do not share records.

The problem to solve is to build **one connected web platform where a company manages its customers and sales pipeline, guarantees that every incoming lead is acted on, and talks to customers, collects payments and schedules appointments through real external services, with every action traceable.**

What goes wrong today

- Leads are assigned informally, so ownership is unclear and responsibility is easy to dodge.
- Absence, leave and overload are not considered when leads are distributed.
- No one is alerted when a lead has been waiting too long, and no one can later prove who was responsible.
- The CRM forces every company into one fixed structure of fields and stages, so teams keep side spreadsheets.
- Customer conversations, payments and appointments live in other apps, so staff switch between tools and the customer record is incomplete.
- Managers cannot see the true state of the pipeline, the response times or the missed opportunities in one place.

What a successful solution achieves

- Every incoming lead has exactly one accountable owner at any moment, and escalates automatically if ignored.
- Each company can shape its own pipeline, fields and rules without developer help.
- Real messages, real payment events and real calendar events appear on the right customer record, with nothing silently lost when an external service fails.
- Access to information depends on the person's role and permissions, enforced on the server.
- Management sees current performance and the history of every important action.

2. Target Users

The platform serves a sales-driven service company with 5 to 30 employees, such as a real estate agency, a solar installer or a clinic's patient-acquisition team. Several companies use the same platform, and each must be completely separate from the others.

User	Situation	What they need
Owner	Runs the company and pays for the platform	Control over the workspace, roles and settings; confidence that no lead is wasted; a clear view of

		revenue and performance
Admin	Delegated by the Owner to configure the system	Ability to customize pipelines, fields, assignment rules and integrations without code
Manager	Supervises a sales or support team	Visibility of who is available, which leads are overdue, and the power to reassign
Sales or support employee (User)	Handles leads and customer conversations daily	One place to see assigned leads, reply to customers, record notes, and get paid or book appointments
Customer (external)	Contacts the company and expects a prompt reply	A fast response on the channel they used, a clear payment link and a confirmed appointment

Owner, Admin and User are the three base levels. A manager is a configurable role built from permissions, not a fourth hard-coded level.

3. Scope

The full long-term vision covers about fifteen business modules. This problem statement defines the part that must be solved now: protecting leads, a customizable CRM, and real connections to external services.

Must be solved	Not part of this problem
Accounts, workspaces, roles and configurable permissions	Full accounting: ledgers, reconciliation, financial statements
Customizable CRM: contacts, companies, opportunities, custom fields, one customizable pipeline	Payroll, shift scheduling and legal leave or break rules
Lead capture, assignment, acknowledgement, escalation and recovery	Advertising-platform analytics and campaign attribution
Audit history of important actions	Full report builder and business intelligence
Operational dashboard on live data	Training and onboarding modules, AI role-play
Real third-party integrations for messaging, payments and calendar, plus one more of the team's choice	Public marketing website, billing for the platform itself

4. Functional Requirements

Each requirement states what must be true in the running application, using real stored data. Timers and thresholds must be configurable so they can be observed in seconds during a demonstration.

F1. Accounts, roles and permissions

ID	Requirement	Acceptance criteria
F1.1	Sign up, log in, log out and persistent sessions	Signing up creates a new company workspace and makes the first user its Owner; refreshing the page keeps the user logged in
F1.2	Three base levels: Owner, Admin, User	The Owner can invite users and assign levels; an Admin cannot modify the Owner; a User sees only what their permissions allow
F1.3	Configurable permissions, not tied to role names	The Owner can grant or remove at least: view, create, edit, delete and export for CRM records, manage assignment rules, manage integrations, and view the audit log
F1.4	Field-level protection	Opportunity value is hidden or masked for users without the matching permission in lists, detail pages and exports
F1.5	Server-side enforcement	A direct API call by an unauthorized user is rejected; hiding a button is not sufficient
F1.6	Workspace isolation	A user of one workspace can never read or change data of another workspace, including through guessed identifiers

F2. Customizable CRM

ID	Requirement	Acceptance criteria
F2.1	Contacts, companies and opportunities with relationships	A contact belongs to a company and can have many opportunities; the contact page shows all related records
F2.2	Custom fields created from the interface	An Admin adds a field with a name and a type (text, number, date, dropdown, checkbox) without code; it appears on forms and detail pages immediately
F2.3	Different fields for different pipelines	Two pipelines show different custom field sets while sharing the same contact information
F2.4	Stage-specific required fields	Moving an opportunity into a configured stage is blocked, with a clear message, until its required fields are filled
F2.5	Customizable pipeline	An Admin can add, rename, reorder and delete stages; deleting a stage that contains opportunities is handled safely; opportunities can be moved

		between stages on a board or list
F2.6	Filtering and saved lists	A user combines at least three criteria (field, tag, stage, owner, date), saves the result as a list, and the list updates automatically as records start or stop matching
F2.7	Activity history	Each contact and opportunity shows a time-ordered log of notes, stage changes, assignments, field edits, messages, payments and appointments, with who and when
F2.8	Complete customer view	One contact page shows conversations, opportunities, payments, appointments, the assigned employee and the activity history

F3. Lead capture, assignment, acknowledgement and escalation

This is the central business problem. A lead must never be left without an accountable person.

ID	Requirement	Acceptance criteria
F3.1	Lead capture from multiple sources	Leads are created from a public web form, from an incoming message on the messaging integration (Section 5, I1) and from a manual entry, each with a source label
F3.2	Employee availability	Each employee has a status (Available, Busy, On leave, Offline) that they or an Admin can set
F3.3	Configurable assignment rules	An Admin configures at least round-robin distribution among Available employees and a maximum number of open leads per employee; employees who are not Available never receive leads
F3.4	Acknowledgement window	The assignee receives a notification and must acknowledge or act within a configurable time; the remaining time is visible
F3.5	Escalation chain	When the window expires, the lead moves to the next eligible employee, then to a designated backup, then to a manager; the number of levels and the time allowed per level are configurable
F3.6	Reassignment on absence	Setting an employee to On leave or Offline immediately reassigns their unacknowledged leads; the treatment of already acknowledged leads is defined and visible to users

F3.7	Recovery queue	One page lists leads that are unacknowledged beyond their window, escalated more than once, or untouched for a configurable period, with the reason shown for each
F3.8	No duplicate action	A lead is never owned by two people at once, and two timers expiring close together do not escalate the same lead twice
F3.9	External notification of escalations	When a lead escalates, the new owner is also alerted outside the application through the integration chosen in Section 5, I4

F4. Audit history

ID	Requirement	Acceptance criteria
F4.1	Record of important actions	Assignment, acknowledgement, escalation, reassignment, role or permission changes, field or pipeline configuration changes, integration connection changes and deletions are recorded with actor (or system), time, target and before and after values where relevant
F4.2	Restricted visibility	Only users with the audit permission can open the history, and it can be filtered by event type, user and date
F4.3	Tamper resistance	No screen or endpoint allows editing or deleting history entries

F5. Operational dashboard

ID	Requirement	Acceptance criteria
F5.1	Live overview	The dashboard shows at least: new leads today, open pipeline value by stage, average time to first response, items in the recovery queue, escalations in the last 24 hours, employee availability, upcoming appointments and payments received this month
F5.2	Consistent definitions	Every figure follows the definitions in Section 7 and matches the underlying records when opened
F5.3	Drill-down	Clicking at least three figures opens the filtered list of records the user is allowed to see
F5.4	Role-aware view	A User sees personal figures (own leads, own response

		time); Owner and Admin see company-wide figures
F5.5	Basic customization	A user can show or hide figures and the choice is remembered

5. Mandatory Third-Party Integrations

A business platform is only useful if it works with the services a company already uses. The solution must connect to **real external services in their sandbox or test modes**, not to fake or hard-coded stand-ins. Three integrations are mandatory (I1 to I3), and a team must choose one more (I4). Each external service has its own authentication scheme, event format, failure behavior and limits, and the solution must handle them correctly.

ID	Integration	Provider (sandbox or test mode)	Business purpose
I1	Messaging	Twilio WhatsApp Sandbox (Twilio SMS is an accepted alternative)	Talk to customers and capture leads from incoming messages
I2	Payments	Stripe in test mode	Collect payments for opportunities and see their status on the customer record
I3	Calendar	Google Calendar through OAuth 2.0	Book appointments with customers on employees' real calendars
I4	Notifications (choose one)	Slack, Telegram bot, or transactional email (for example Resend, SendGrid or Mailgun)	Alert employees outside the application when a lead escalates

I1. Messaging

ID	Requirement	Acceptance criteria
I1.1	Receive messages	A message sent from a real phone to the sandbox number reaches the platform through the provider's webhook and appears in a conversation view
I1.2	Request authenticity	Webhook requests are verified with the provider's signature mechanism; a forged or altered request is rejected and recorded
I1.3	Send messages	An employee replies from the platform and the customer receives it on their phone
I1.4	Delivery status	Sent, delivered and failed states from the provider's status callbacks are stored and shown on each message
I1.5	Contact matching	An incoming message from a known number attaches to the existing contact; from an unknown number it creates a contact and a lead (F3.1)

I1.6	Ownership	Each conversation shows its assigned employee and is visible only to users permitted to see it
I1.7	Failure handling	If the provider rejects a send or is unreachable, the message stays visible as failed with the reason, and the employee can retry; nothing is silently dropped
I1.8	Duplicates	A webhook delivered twice by the provider does not create two messages or two leads
I1.9	Opt-out	A contact marked as opted out cannot be messaged, and the interface explains why

I2. Payments

ID	Requirement	Acceptance criteria
I2.1	Create a payment request	From an opportunity, an employee creates a payment link or checkout session for a stated amount and currency, and can send it to the customer
I2.2	Payment confirmation by webhook	Completing a test payment updates the opportunity's payment status in the platform without any manual refresh or action, driven by the provider's webhook
I2.3	Request authenticity	Payment webhooks are verified with the provider's signature mechanism before any record changes
I2.4	Idempotent processing	The same event received more than once records the payment once
I2.5	Other outcomes	Expired, abandoned and failed payments are shown with the correct status; a refund made in the provider's dashboard is reflected in the platform
I2.6	Collected versus expected	The platform distinguishes collected revenue from projected or unpaid amounts, and the dashboard (F5.1) uses collected revenue only for payments received
I2.7	Traceability	Each payment shows the opportunity, contact, employee who requested it and the provider's reference
I2.8	No card data stored	The platform never stores or logs card numbers; payment details are entered only on the provider's hosted page

I3. Calendar

ID	Requirement	Acceptance criteria
I3.1	Connect with consent	An employee connects their own Google account through the OAuth 2.0 consent flow, and can disconnect it
I3.2	Token handling	Access tokens are refreshed automatically; tokens are stored only on the server; a revoked or expired connection is detected and shown as needing reconnection
I3.3	Create appointments	From an opportunity or contact, an employee books an appointment that appears in the employee's real Google Calendar and on the customer record
I3.4	Avoid conflicts	The booking screen reads the employee's busy times and prevents or warns about overlaps
I3.5	Time zones	Appointment times display correctly for the employee and are stored unambiguously, including across different time zones
I3.6	Change and cancel	Rescheduling or cancelling in the platform updates the Google event, and the change is recorded in the activity history
I3.7	Availability link	Calendar connection status is visible to Admins; an employee without a connected calendar cannot be booked and sees why

I4. Notifications (choose one provider)

ID	Requirement	Acceptance criteria
I4.1	Real delivery	When a lead escalates (F3.5), the new owner receives a message on the chosen channel within the same minute
I4.2	Useful content	The alert names the lead, the source, the reason for escalation and a link that opens the lead
I4.3	Per-user routing	Each employee can configure their own destination (Slack user, Telegram chat or email address)
I4.4	Failure handling	A failed alert is recorded, retried a limited number of times and visible to Admins

I5. Requirements shared by every integration

ID	Requirement	Acceptance criteria

I5.1	Central integrations page	One page lists each integration with its status (not connected, connected, error), when it last worked and its recent errors
I5.2	Per-workspace configuration	Each workspace supplies its own credentials or connection; one workspace's connection is never used by another
I5.3	Secret handling	API keys, tokens and signing secrets are never sent to the browser, never committed to the repository and never shown in full in the interface
I5.4	Synchronization history	Every inbound event and outbound call is recorded with time, result and a reference, and can be reviewed by authorized users
I5.5	Resilience	Temporary provider errors are retried with a delay; permanent errors are shown to a user with a plain explanation
I5.6	Clearly stated limits	The interface states what each integration can and cannot do (for example, that the messaging sandbox only reaches phones that have joined it)
I5.7	Permission and audit	Only users with the integration permission can connect, change or remove an integration, and each change is recorded in the audit history (F4.1)
I5.8	Isolation of failures	If one provider is unavailable, the rest of the platform keeps working

Provider constraints to plan for

These are the practical conditions of the chosen providers. Confirm them against each provider's current documentation, since they can change.

Provider	What is needed	What to expect
Twilio WhatsApp Sandbox	A Twilio account and the sandbox number	A phone must first join the sandbox by sending the join phrase; sandbox messages are for testing only
Stripe test mode	A Stripe account in test mode, no business verification	Payments use published test card numbers; webhook secrets are issued per endpoint
Google Calendar	A Google Cloud project, enabled Calendar API and an OAuth consent screen	An app in testing status accepts only listed test users and may issue short-lived refresh tokens

Slack, Telegram or email provider	A free account or bot with its own token	Each has its own message format, rate limits and error codes
All webhooks	A publicly reachable HTTPS address	A local development setup needs a tunnel or a deployed address for providers to reach it

6. Quality Requirements

These requirements apply to everything in Sections 4 and 5. They describe how well the platform must behave, and they are where core web development skill shows.

ID	Area	Requirement
Q1	Authentication security	Passwords are stored only as salted hashes from a recognized algorithm; sessions or tokens expire and can be ended; login is protected against repeated guessing
Q2	Authorization	Every endpoint checks the caller's workspace and permission on the server; identifiers in URLs cannot be used to reach other users' or other workspaces' data
Q3	Input safety	All input is validated on the server; the application is protected against injection, cross-site scripting and cross-site request forgery; uploaded or imported content is never trusted
Q4	Data integrity	Related records stay consistent: deleting or changing a stage, field, user or contact never leaves orphaned or contradictory data; concurrent edits do not silently overwrite each other
Q5	Concurrency and timing	Assignment, escalation and webhook handling stay correct when events arrive at the same time or more than once, and when the server restarts while timers are running
Q6	Persistence	All data lives in a real database and survives restart; configuration made through the interface is stored per workspace
Q7	Responsive interface	Every screen is usable on a phone, a tablet and a laptop without horizontal scrolling or hidden controls
Q8	Accessibility	Interactive elements can be used with a keyboard, forms have labels, color is not the only carrier of meaning, and contrast is readable

Q9	Performance	Lists with several hundred records load with pagination or equivalent, search and filters respond without freezing the interface, and slow operations show progress while the user keeps working
Q10	Feedback and errors	Every action shows loading, success and failure states; error messages say what went wrong and what the user can do
Q11	Real-time awareness	New leads, escalations and incoming messages appear without a manual page refresh
Q12	Consistent data	The same figure shows the same value on every screen, because all screens read the same records and definitions
Q13	Configuration and secrets	Settings and secrets are kept outside the source code; a new person can start the application from written instructions and sample data
Q14	Maintainable structure	The code separates business rules, data access, integrations and presentation so that a provider or rule can change without rewriting the application

7. Glossary and Metric Definitions

Each term below has one meaning everywhere in the platform: on the dashboard, in lists, in exports and in the audit history.

Term	Definition
Workspace	One company's isolated environment, with its own users, data, configuration and integrations
Lead	A record created by intake (web form, incoming message or manual entry) with a source label
Assigned time	The moment the system last assigned the lead to a person
Acknowledged	The assignee clicked Acknowledge, or performed a recorded action on the lead (note, message, call log, stage change, appointment)
Time to first response	Time from the lead's first assignment to its first acknowledgement
Escalation	An automatic reassignment caused by an expired acknowledgement window; manual reassignments are counted separately
Recovery queue item	A lead that is unacknowledged beyond its window, has escalated

	two or more times, or has no recorded action for the configured idle period
Available employee	A user whose status is Available and whose open-lead count is below the configured limit
Open pipeline value	The sum of values of opportunities not in a won or lost stage, grouped by stage; shown only to users permitted to see values
Projected revenue	The value of open opportunities and of payment requests not yet paid
Collected revenue	The sum of payments confirmed by the payment provider's webhook, minus confirmed refunds
Conversation	All messages with one contact on one channel, linked to the contact, its assigned employee and, when applicable, an opportunity
Appointment	A calendar event created from the platform and linked to a contact or opportunity
Integration event	Any inbound webhook or outbound call to an external provider, recorded in the synchronization history